# PHYS 101 - Fundamentals of Physics I
# Fall Quarter 2026-2027

---

**Faculty**

**Course Director and Section A Lecturer**
Dr. Bárbara Cruvinel Santiago (barbara.cruvinelsantiago@drexel.edu)
Office: Disque 810
Office Hours: Tuesdays 1:30 – 3 pm and Thursdays 11 am – 12:30 pm, or by appointment

**Teaching Assistants**

**Quinn Cooney** (qc78@drexel.edu)
Office Hours: Mondays 3 – 5 pm (Korman 103)

**Bryan Kim** (bak333@drexel.edu)
Office Hours: Tuesdays 1 – 3 pm (Korman 103)

**Course time and location:** Tuesdays 10 am – 11:50 am (Disque 108)

**Course Canvas Page** (https://drexel.instructure.com/courses/11919)
Links to lecture resources, recitation assignments, homework assignments, conceptual exercise assignments, laboratory experiments, and other course resources are accessed on the PHYS 101 Canvas page.

**Course Materials**

**Textbook:** "*Physics for Scientists and Engineers - 6th Edition"* by Tipler and Mosca
**Macmillan Achieve Access:** https://achieve.macmillanlearning.com
You must purchase an Achieve book/homework package before enrolling in our online homework course. The Drexel Bookstore and Macmillan both have options for Achieve standalone access with an e-book and for bundles with physical copies of the textbook that come with an access code. Make sure that your numerical Drexel address is signed up with Achieve. Then go to Canvas and link your Canvas and Achieve accounts by clicking on the Macmillan link available under Modules on Canvas. If you want to check that you're in the correct course, our Homework Course ID is **te9ecv**.

**Course Description** (Catalog link)
First of a three-course physics sequence introducing mechanics to engineering and science majors. Topics include: Translational kinematics of particles in one and two dimensions. Inertial and non-inertial frames, Newton's laws of motion, force, energy and momentum. Concepts of conservation of energy and momentum. Rotational kinematics and dynamics.

**Course Learning Outcomes**

- Define key concepts of the fundamental physics laws in mechanics: the concept of motion and Newton's laws, conservation of energy, linear momentum and angular momentum.
- Apply fundamental laws of physics to develop approaches to solve physics problems in topics in classical mechanics.
- Apply conservation laws to analyze and correctly predict outcomes for various physical situations involving classical mechanics.
- Perform laboratory experiments and evaluate results including uncertainties to draw conclusions with relation to theoretical predictions.
- Develop analytic skills toward analysis and problem solving that can be applied in science and engineering disciplines.

**Course Organization**

***Lectures*** will introduce concepts and present examples. See *Course Attendance and Participation* for information on in-class assignments.

***Recitations*** will discuss concepts and solve problems using principles learned in lectures. You'll turn in a problem at the end of every recitation that will count for your recitation participation grade.

***Laboratories*** will explore ideas and concepts in hands-on, team-oriented activities

Four **Laboratory Experiments** will be conducted during the term to relate observation to theory, to model physical systems, and to practice data analysis. Pre-Lab Videos, laboratory procedures, and worksheets are posted on the Canvas course page under Files. *Watching* ***Pre-Lab Videos*** *prior to the laboratory meeting is expected.* A Laboratory Worksheet will be completed by each lab group, and a single grade will be given to each Lab Group report.

Lab worksheets are due exactly a week after your lab section. Late worksheets will lose 20% in their grade for every day that it is late. Only one student needs to submit the worksheet per group and should select the names of all group members on Canvas when doing so to ensure that everyone gets a grade.

Lab sessions begin in **Week 2**
Even lab sections (60, 62, 64, 66) will meet in even class weeks (2, 4, 6 and 8)
Odd lab sections (61, 63) will meet in odd class weeks (3, 5, 7 and 9)

| | |
|---|---|
| **Laboratory 1** | **Measurement and Uncertainty in Scientific Experiments** |
| **Laboratory 2** | **Kinematics of One-Dimensional Motion: Motion of a Cart on a Ramp** |
| **Laboratory 3** | **The Simple Pendulum** |
| **Laboratory 4** | **Conservation of Momentum in One-Dimension** |

***Homework:*** **Achieve** includes required assignments and is accessible Canvas under Modules. Note that each week assigned material on the Achieve website will include some helpful material that is not worth a grade (but that should help you studying and understanding concepts!), a set of conceptual exercises that is worth 50 points and a set of problems that is worth 100 points. Problems will be available 7 days before they are due (the last homework will be available 2 weeks in advance). Submissions are due on Mondays at 11:59 pm.

Exceptions to the homework deadlines are noted on the course calendar on this syllabus.

Late homework will receive a penalty of 20% by day that it's late.

***Examinations***
Two one-hour **Midterm Examinations** and a comprehensive two-hour **Final Examination** will be given. A make-up examination* may be arranged if special circumstances arise with the permission of the Course Director. Exam locations will be listed on Canvas.

* Conditions for permission for Make-up Examination can be found under Course Attendance and Participation

| **Midterm** | **Day/Date** |
|---|---|
| 1 | **Monday, 10/26/26, 8:00am** |
| 2 | **Monday, 11/9/26, 8:00am** |
| Make-up* | **Monday, 11/23/26, 8:00am** |

**Final Examination** – Date, time and location to be announced on Canvas

- A scientific calculator without internet connection is required for all examinations.
- An equation sheet will be released on Canvas prior to each examination for review and printed with each examination.

**Syllabus Note**
Minor changes may be made to the Syllabus over the course of the term. These will be announced in lecture when possible and always posted on the course Canvas page.

**Grading:** The final course grade will be based on the following weights.

| | |
|---|---|
| Lecture Attendance and Participation | 5% |
| Recitation Attendance and Participation | 10% |
| Achieve (Conceptual Exercises and Set of Problems) | 15% |
| Laboratories (4) | 20% |
| Midterms (2) | 30% |
| Final Exam | 20% |
| **TOTAL** | **100%** |

**Letter Grades**

| **Course numerical grades** | **Letter Grade** |
|---|---|
| Score > 96 % | A+ |
| 90% < Score < 96% | A |
| 86% < Score < 90% | B+ |
| 80% < Score < 86% | B |
| 76% < Score < 80% | C+ |
| 70% < Score < 76% | C |
| 65% < Score < 70% | D+ |
| 60% < Score < 65% | D |
| Score < 60% | F |

*The grading scale may be modified towards lower numerical grade thresholds (in students' favor) if needed.*

**Grade Appeals**
If a student has any comments, concerns, or questions about provided class materials and/or recording, speak with your course instructor first. If this does not resolve the issue, you can also reach out to the Department Head, and use the process described for a grade appeal at Grade Appeals.

Any concerns regarding assigned scores on *any* graded assignments must be brought to the instructors' attention within one calendar week of the assignment's due date. In no event will re-scoring for or adjustments to online assignments be considered after the last regular day of classes.

**Course Attendance and Participation**
Attendance at and participation in lectures, recitation, laboratory activities and exams is mandatory, and will be recorded.

Attendance in lecture will be recorded with simple Poll Everywhere problems starting in the second week of classes. These questions be graded as 90% participation and 10% for correctness. Every lecture, you will receive a daily average for those problems, and the average of these daily averages will make up the 5% lecture attendance and participation grade. You can have up to two daily averages of zero dropped for unexcused absences. Beyond that, daily scores of zero will only be dropped for excused absences similarly to the policy for make-up labs and exams (see below). Instructions for how to access Poll Everywhere will be

posted on Canvas on the first week of classes. It's expected that students will bring some device that can connect to the internet to use Poll Everywhere (for example a phone, tablet, or computer).

In some instances, a student may be unable to attend a course activity such as exams, labs and recitations. In such a situation, e-mail and/or discuss the situation with the Course Director before an expected absence or within one week after an unexpected absence to be considered for an excused absence (unless the student is incapacitated and unable to communicate during that period, in which case communication should take place as soon as possible thereafter). Students should refer to the Absence from Class policy that includes the University policy on excused absences including absences for religious observances.

Make-up exams and labs, as well as excused absences in lecture and recitation will only be granted under extenuating circumstances including but not limited to illness, emergencies, varsity trips, religious accommodations. Private details related to health and family issues, for example, do not need to be shared with the instructor/course director unless the student is comfortable and willing to do so.

In case a student has an official flexible attendance accommodation due to a disability, arrangements will be made around the course policy accordingly to honor such accommodations.

**Disability Access**
Students with disabilities requesting accommodations must present a current Accommodation Verification Letter (AVL) to the Course Director before accommodations can be made. AVLs are issued by the Disability Resources Team. Resources for students with disabilities can be found at the following website.

Disability Resources
Registering with Disability Resources

**Tutoring and Student Resources**
The Physics Help! Center offers free tutoring sessions to Drexel students in introductory physics courses. Consult the Physics Help! Center calendar for tutoring schedules.

Additional student resources and academic support can be found at
https://drexel.edu/coas/academics/student-resources-support

Help with technology can be found on the Information Technology website at
https://drexel.edu/it

**Health Center and Student Counseling**
Health and wellness resources for students can be found at the following websites.

http://drexel.edu/counselingandhealth/student-health-center/overview
http://drexel.edu/counselingandhealth/counseling-center/overview

**Academic Integrity**
Collaboration with classmates to study and work on homework sets is encouraged, but the final solution should always be your own. While asking AI services for clarification on concepts is allowed, asking AI to solve problems for you in any activities or labs is also forbidden. Breaches of academic integrity, for example, cheating or plagiarizing from another student or copying from online sources without reference, are serious infractions of the community standards of Drexel University. Likewise, breaches of computer systems policies serious infractions of the community standards of Drexel University. Refer to the following websites to understand expectations of students and consequences for violating these standards.

Academic Integrity
Student Code of Conduct
Security of Enterprise Systems
Security of Information and Networked Systems

**Commitment to a welcoming, inclusive and respectful environment**
The PHYS 101 instructors recognize the talents and diversity of all of our students. Students' perspectives benefit the learning process and the entire Drexel University community. We will strive to be sure that materials and activities in this course respect the diverse backgrounds of our students including race, ethnicity, gender, sexuality, disability, age, religion, socioeconomic status, and culture. We believe that each student should be respected by us and by their peers. Please feel free to approach the instructors with any suggestion on how to make our classroom environment more welcoming.

Civil Rights Compliance

**Appropriate Use of Course Materials**
It is important to recognize that some or all of the course materials provided to you are the intellectual property of Drexel University, the course instructor, or others. Use of this intellectual property is governed by Drexel University policies, including the IT-1 policy found at Acceptable Use Policy. Briefly, this policy states that **all course materials including recordings** provided by the course instructor may not be copied, reproduced, distributed or re-posted, unless otherwise given prior written approval by the University. Doing so may be considered a breach of this policy and will be investigated and addressed as possible academic dishonesty, among other potential violations. Improper use of such materials may also constitute a violation of the Code of Conduct.

**Recording of Class Activities**
In general, students and others should not record course interactions and course activities in lecture, lab, or recitation. Students who have an approved accommodation from the Office of Disability Resources to record online lectures and discussions for note taking purposes should inform their course instructor(s) of their approved accommodation in advance. The recording of lectures and discussions may only be carried out by the students enrolled in the class who have an approved accommodation from Disability Resources with their instructors' prior knowledge and consent. Students with approved accommodations may be asked to turn off their recorder if confidential or personal information is presented.

**Registration and Grading Policies**
Please note the following policies for adding, dropping, or withdrawing from courses at Drexel University as well as incomplete grades.

Course Add/Drop
Course Withdrawal
Incomplete Grades

**Course Schedule**

| Week # (Lecture Day) | Anticipated lecture topics | Labs | Homework (due by 11:59 pm on the day listed) |
|---|---|---|---|
| 1 (Sep 22) | Chapter 1 | No labs | Monday, September 28, 2026 |
| 2 (Sep 29) | Chapter 2 and Chapter 3 (partial) | Lab 1 for even sections | Monday, October 5, 2026 |
| 3 (Oct 6) | Chapter 3 (partial) and Chapter 4 | Lab 1 for odd sections | Monday, October 12, 2026 |

| | | | |
|---|---|---|---|
| 4 (Oct 13) | Chapter 5 | Lab 2 for even sections | Monday, October 19, 2026 |
| 5 (Oct 20) | Chapter 6 and Chapter 7 (partial) | Lab 2 for odd sections | Monday, October 26, 2026 |
| 6 (Oct 27) | Chapter 7 (partial) and Chapter 8 (partial) | Lab 3 for even sections | Monday, November 2, 2026 |
| 7 (Nov 3) | No lecture due to Election Day (go vote!), but recitations and labs still meet | Lab 3 for odd sections | No homework! (Utilize this week to study for the second exam since there's no class ) |
| 8 (Nov 10) | Chapter 8 (partial) and Chapter 9 (partial) | Lab 4 for even sections | Monday, November 16, 2026 |
| 9 (Nov 17) | Chapter 9 (partial) and Chapter 10 (partial) | Lab 4 for odd sections | Monday, November 23, 2026 |
| 10 (Nov 24) | Chapter 10 (partial) | No recitations or labs (Thanksgiving Break) | No homework! (Thanksgiving Break) |
| 11 (Dec 1) | Chapter 14 (partial) and review of course topics | Make-up labs for those who qualify for one | Saturday, December 5, 2026 |